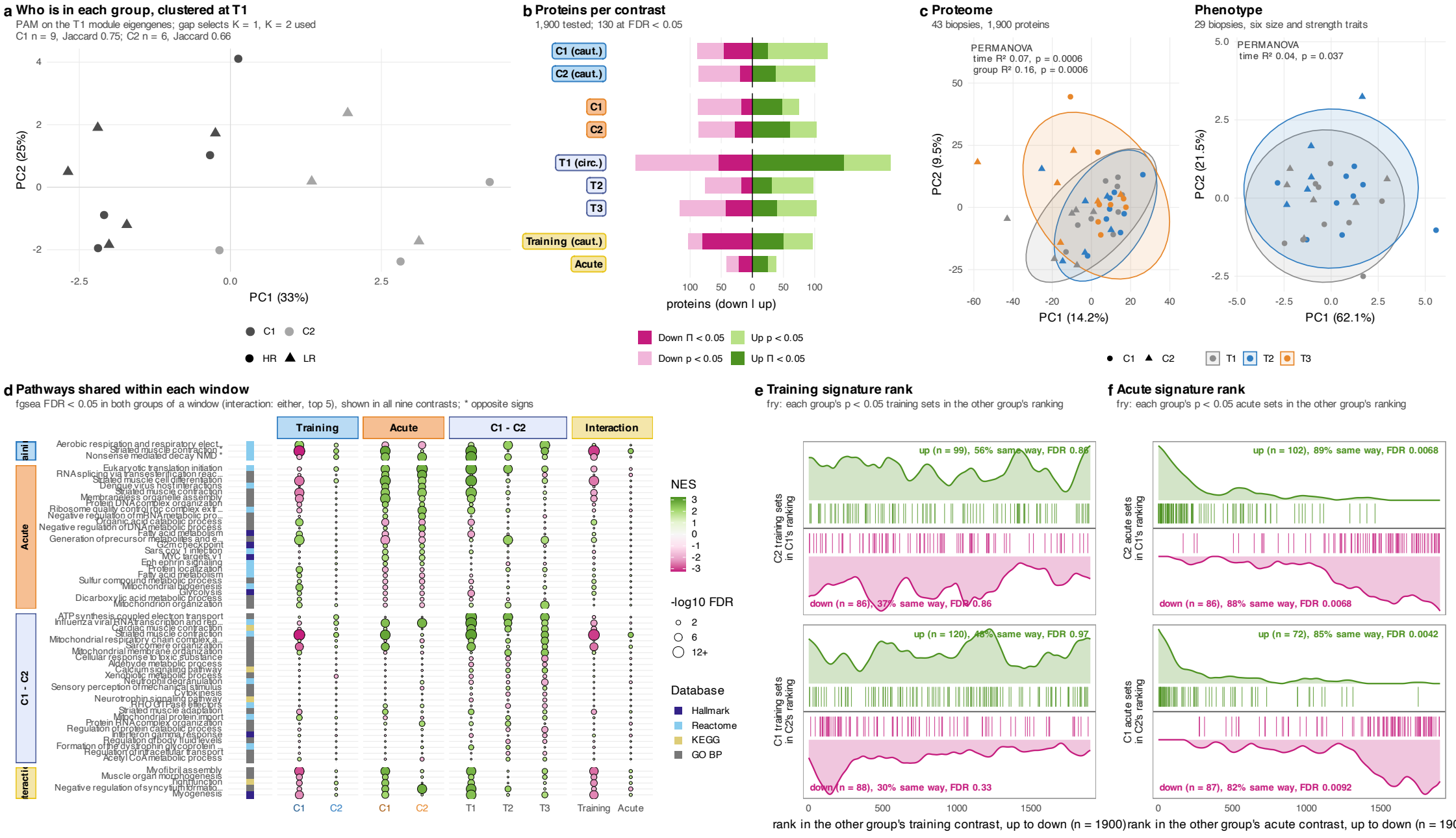

a Who is in each group, clustered at T1
PAM on the T1 module eigengenes; gap selects K = 1, K = 2 used
C1 n = 9, Jaccard 0.75; C2 n = 6, Jaccard 0.66
PC2 (25%)
PC1 (33%)
C1 C2
HR LR
b Proteins per contrast
1,900 tested; 130 at FDR < 0.05
C1 (caut.)
C2 (caut.)
C1
C2
T1 (circ.)
T2
T3
Training (caut.)
Acute
proteins (down | up)
Down Π < 0.05
Up p < 0.05
Down p < 0.05
Up Π < 0.05
c Proteome
43 biopsies, 1,900 proteins
PERMANOVA
time R² 0.07, p = 0.0006
group R² 0.16, p = 0.0006
PC2 (9.5%)
PC1 (14.2%)
Phenotype
29 biopsies, six size and strength traits
PERMANOVA
time R² 0.04, p = 0.037
PC2 (21.5%)
PC1 (62.1%)
C1 C2
T1 T2 T3
d Pathways shared within each window
fgsea FDR < 0.05 in both groups of a window (interaction: either, top 5), shown in all nine contrasts; * opposite signs
Training
Acute
C1 - C2
Interaction
Acute
C1 - C2
Aerobic respiration and respiratory elect...
Striated muscle contraction *
Nonsense mediated decay NMD *
Eukaryotic translation initiation
RNA splicing via transesterification reac...
Striated muscle cell differentiation
Dengue virus host interactions
Striated muscle contraction
Membraneless organelle assembly
Protein DNA complex organization
Ribosome quality control ribosome ext...
Negative regulation of mRNA metabolic pro...
Organic acid catabolic process
Negative regulation of DNA metabolic process
Fatty acid metabolism
Generation of precursor metabolites and e...
G2M checkpoint
Sars cov 1 infection
MYC targets v1
Eph ephrin signaling
Protein localization
Fatty acid metabolism
Sulfur compound metabolic process
Mitochondrial biogenesis
Glycolysis
Dicarboxylic acid metabolic process
Mitochondrion organization
ATP synthesis coupled electron transport
Influenza viral RNA transcription and rep...
Cardiac muscle contraction
Striated muscle contraction
Mitochondrial respiratory chain complex a...
Sarcomere organization
Mitochondrial membrane organization
Cellular response to toxic substance
Aldehyde metabolic process
Calcium signaling pathway
Xenobiotic metabolic process
Neutrophil degranulation
Sensory perception of mechanical stimulus
Cytokinesis
Neurotrophin signaling pathway
RHO GTPase Effectors
Striated muscle adaptation
Mitochondrial protein import
Protein DNA complex organization
Regulation of protein catabolic process
Interferon gamma response
Regulation of body fluid levels
Formation of the dystrophin glycoprotein ...
Regulation of intracellular transport
Acetyl CoA metabolic process
Myofibril assembly
Muscle organ morphogenesis
Tight junction
Negative regulation of syncytium formatio...
Myogenesis
C1 C2 C1 C2 T1 T2 T3 Training Acute
NES
3 2 1 0 -1 -2 -3
-log10 FDR
2 6 12+
Database
Hallmark
Reactome
KEGG
GO BP
e Training signature rank
fry: each group's p < 0.05 training sets in the other group's ranking
up (n = 99), 56% same way, FDR 0.86
down (n = 86), 37% same way, FDR 0.86
up (n = 120), 46% same way, FDR 0.97
down (n = 88), 30% same way, FDR 0.33
C2 training sets in C1's ranking
C1 training sets in C2's ranking
0 500 1000 1500
rank in the other group's training contrast, up to down (n = 1900)
f Acute signature rank
fry: each group's p < 0.05 acute sets in the other group's ranking
up (n = 102), 89% same way, FDR 0.0068
down (n = 86), 88% same way, FDR 0.0068
up (n = 72), 85% same way, FDR 0.0042
down (n = 87), 82% same way, FDR 0.0092
C2 acute sets in C1's ranking
C1 acute sets in C2's ranking
0 500 1000 1500
rank in the other group's acute contrast, up to down (n = 190